

A hybrid model combining variational mode decomposition and an attention-GRU network for stock price index forecasting

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In this paper we introduce a new hybrid model based on variational mode decomposition (VMD) and Gated Recurrent Units (GRU) network improved by attention mechanism to enhance the accuracy of stock price indices forecasting. In the process of establishing the model, VMD is made a use to decompose the primary series into some almost orthogonal subsequences. The attention mechanism is introduced into GRU to assign different weights to the input elements in advance so that better predictive results can be achieved for each component. In empirical experiment, London FTSE Index (FTSE) and Nasdaq Index (IXIC) are adopted to examine the performance of VMD-AttGRU model. Empirical results report that the developed hybrid model outperforms the single models and indeed raises the accuracy of stock price indices forecasting. In addition, the introduction of attention mechanism can increase the level predictive accuracy but decrease the correctness of direction forecasting.

Keywords: variational mode decomposition, Gated Recurrent Units, attention mechanism, forecasting, stock price

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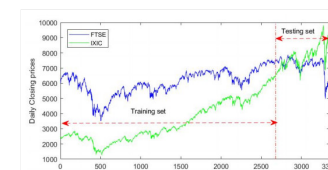
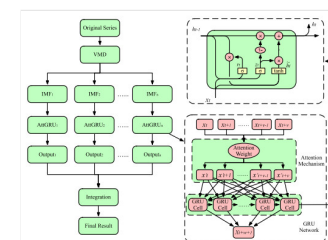
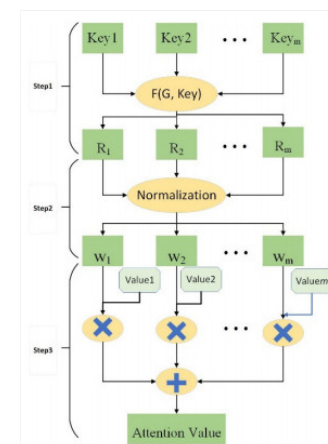
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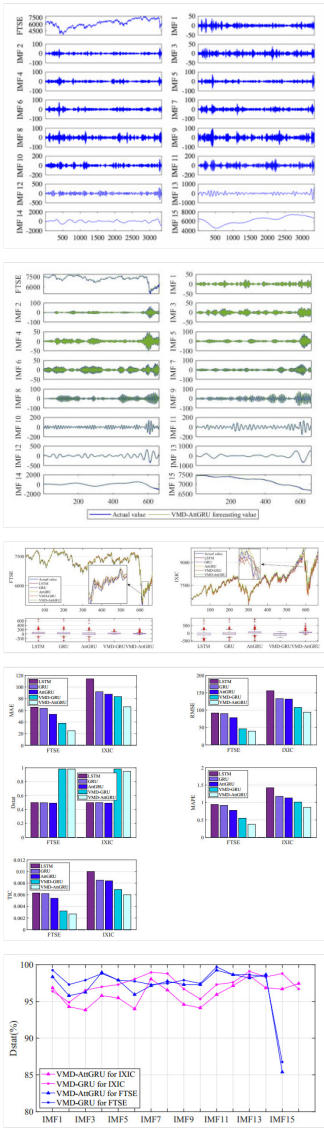
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